

History

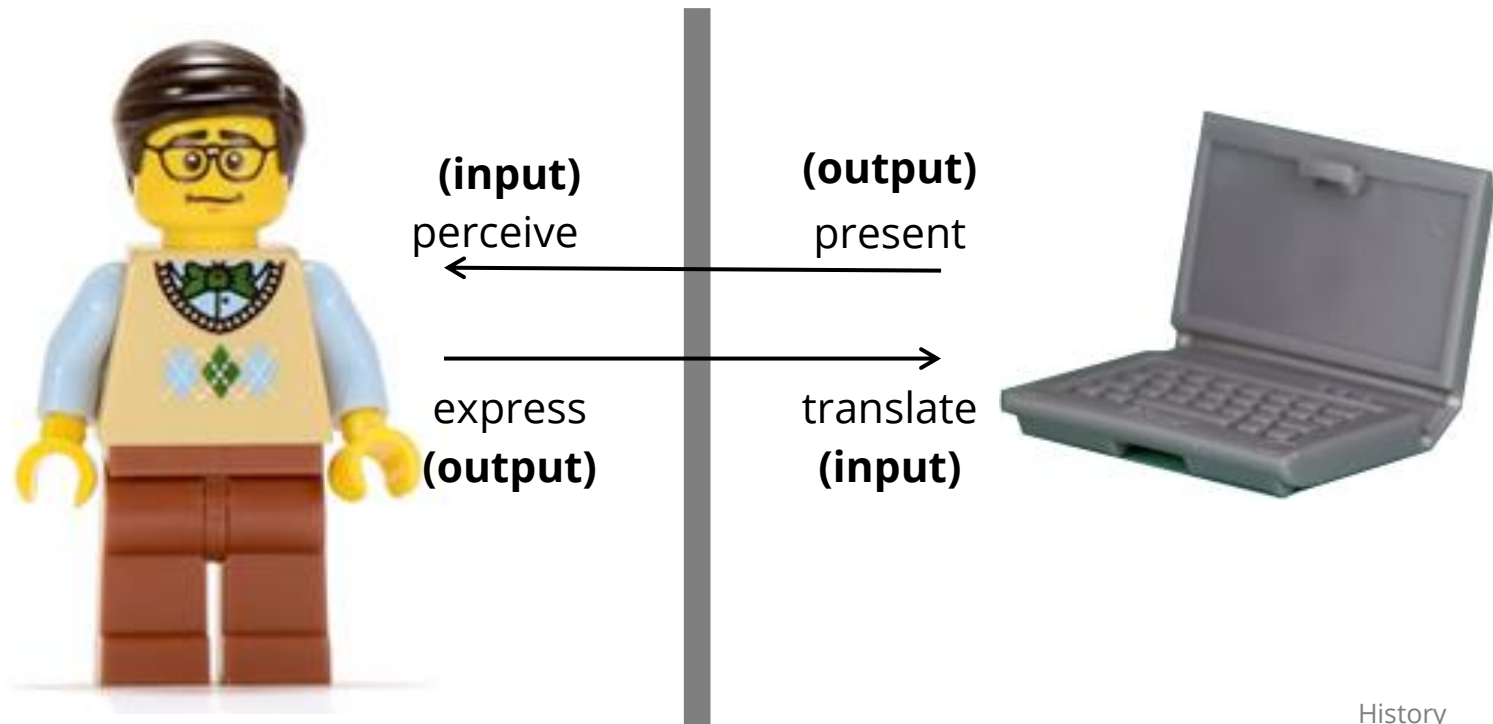
A (brief) history of interaction

Outline

- Major paradigms of interaction
 - Batch interfaces
 - Conversational interfaces
 - Graphical interfaces
- Visionaries who inspired advances
 - Vannevar Bush
 - Douglas Engelbart
 - Ivan Sutherland
 - Alan Kay
- The current and future of interaction

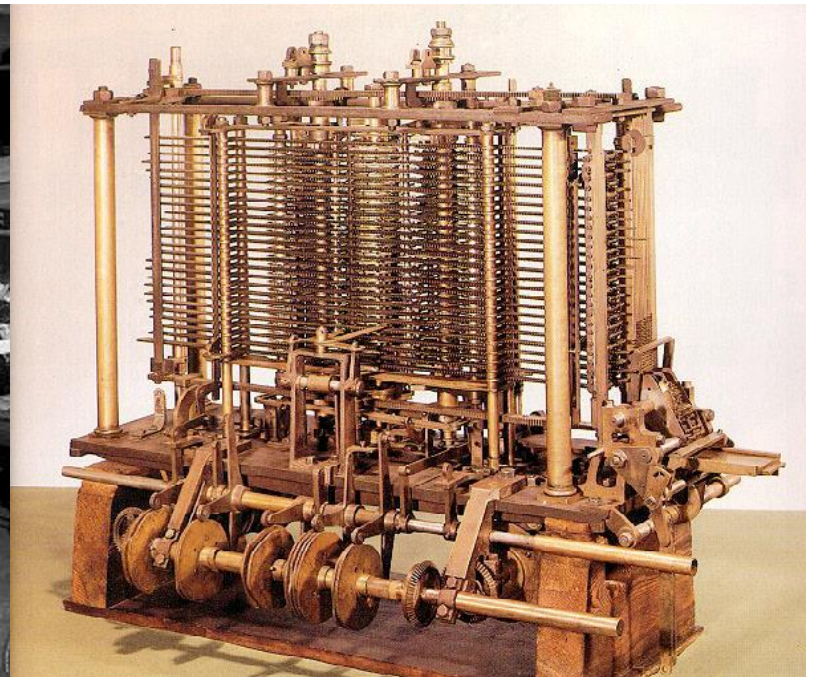
The History of Interaction

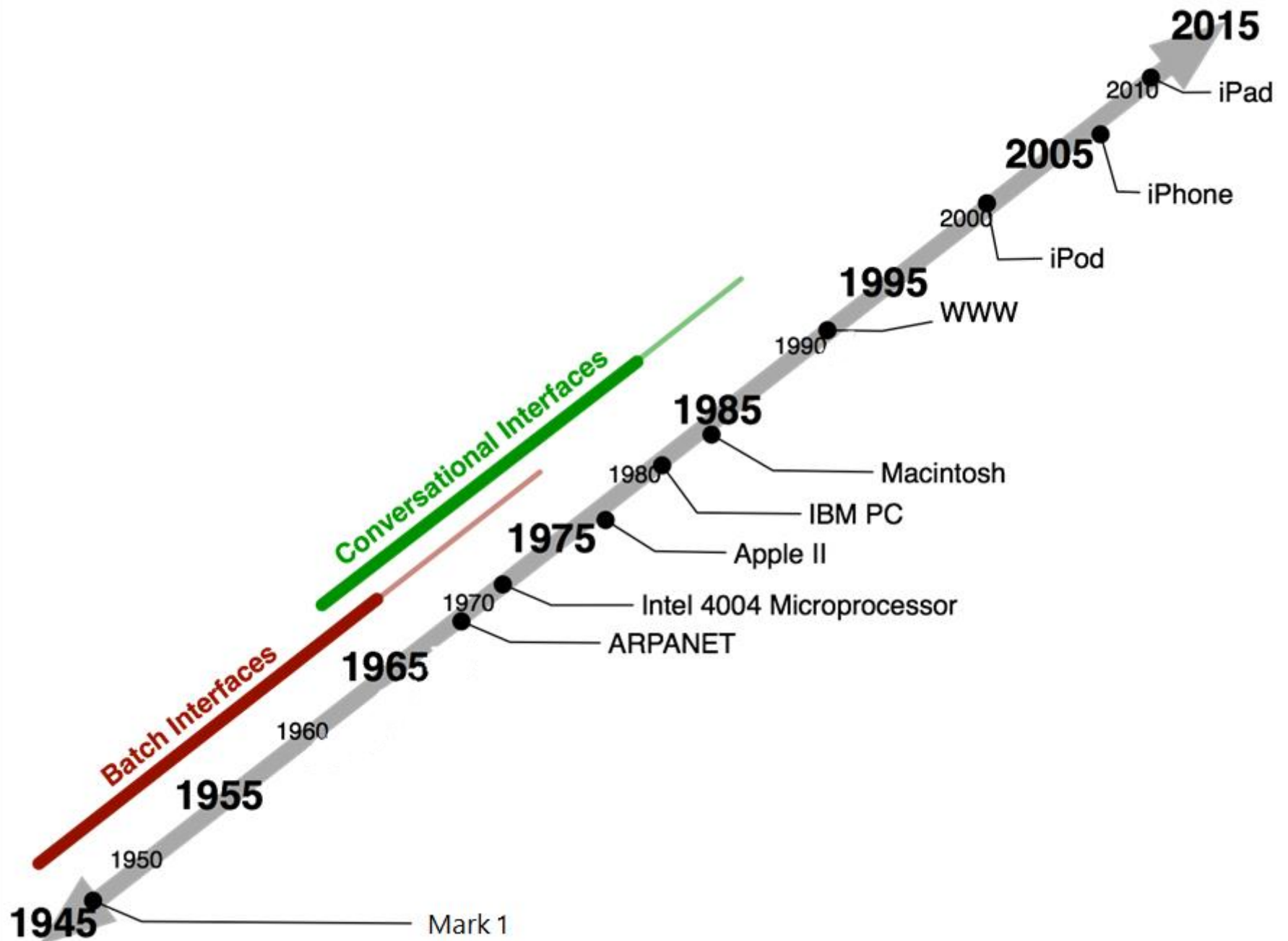
- History of interaction is the history of making the input and output languages of the machine closer to the input and output language of the user and their tasks
- Interaction has evolved from forms that favoured the machine (when its time was more valuable) to those that favour the user



Earliest “Computers”

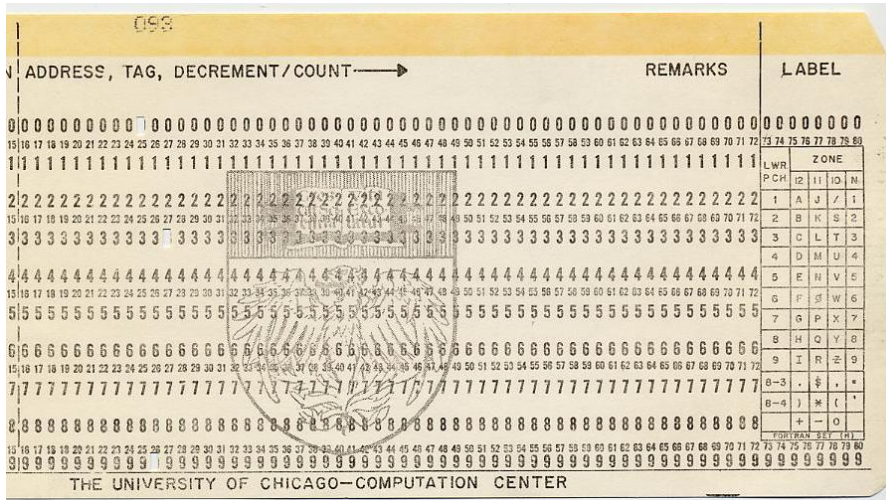
- Human computers (up to 1940s)
- Babbage’s Analytical Engine (designed mid 1800s)
 - First general-purpose computing device
 - Calculate mathematical formulas
 - Promised to do the work of (human) computers much more quickly and accurately



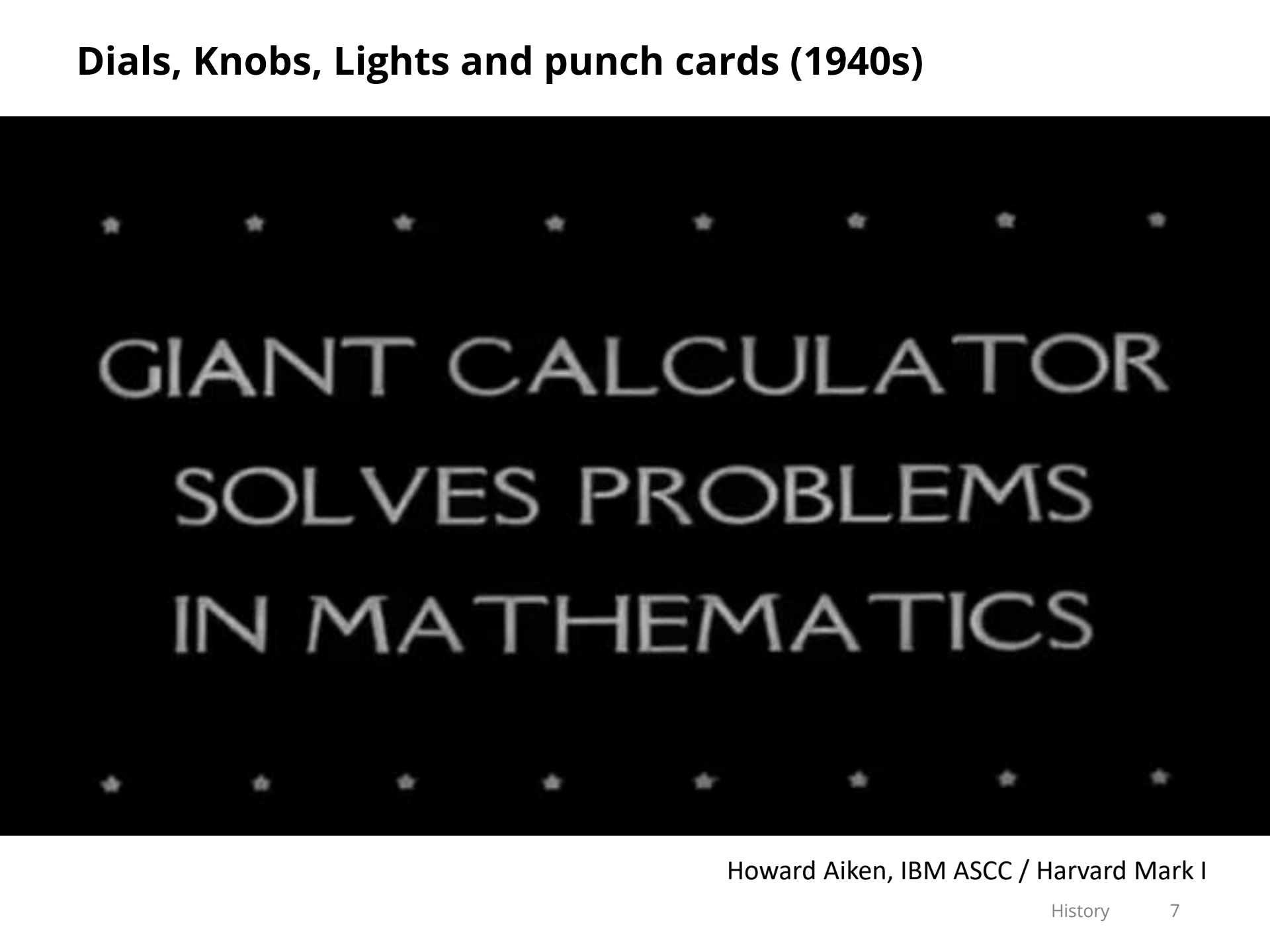


Batch Interface (1945-1965)

- Interaction style
 - Set of instructions prepared a priori
 - Input fed to computer via punch cards, paper tape, magnetic tape
 - Output - typically received via paper printout
 - No real interaction possible as system executes instructions
 - Responses received in hours, days
- Users
 - Only used by highly trained individuals

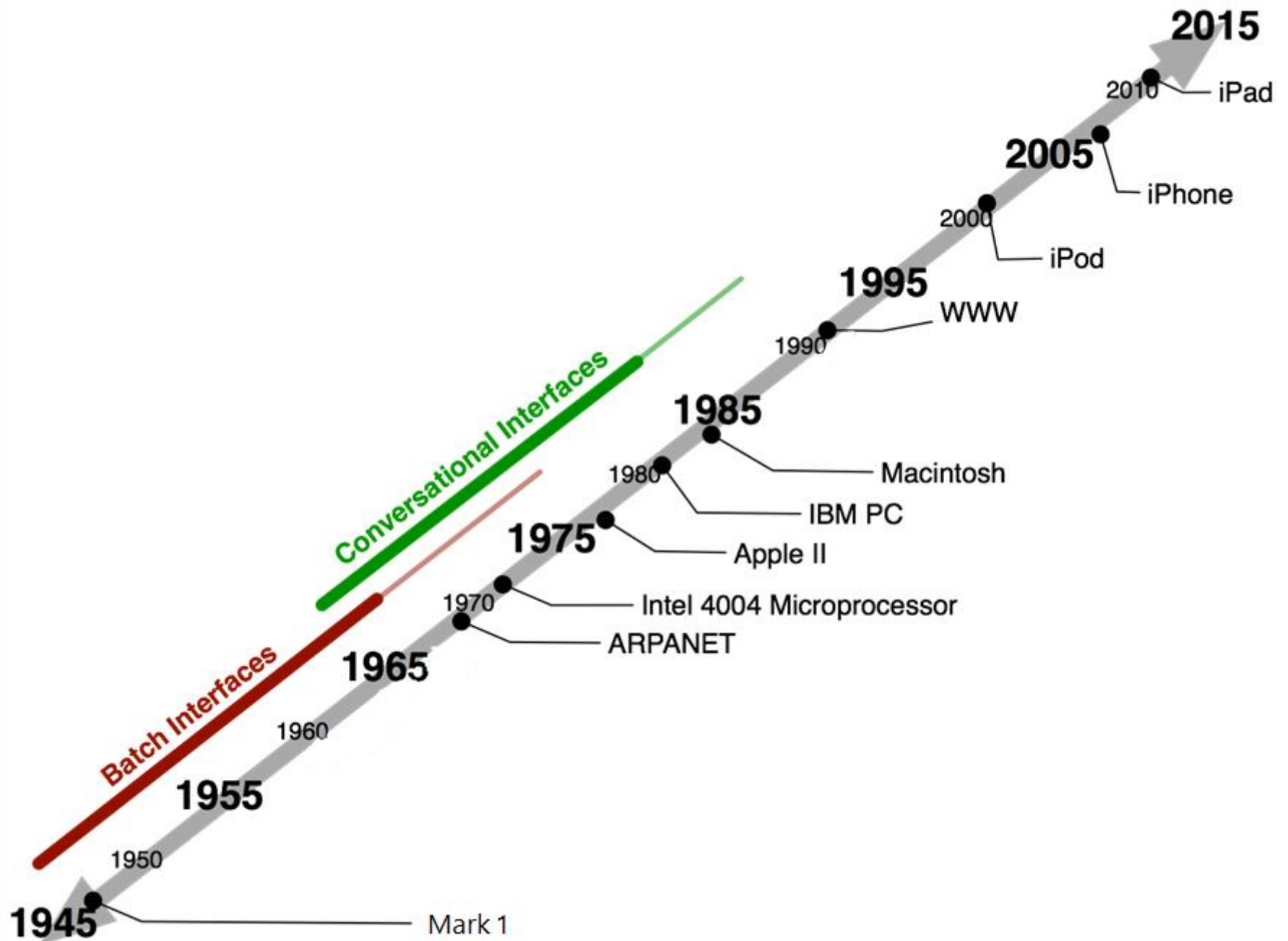


Dials, Knobs, Lights and punch cards (1940s)



GIANT CALCULATOR
SOLVES PROBLEMS
IN MATHEMATICS

Howard Aiken, IBM ASCC / Harvard Mark I



Conversational Interface (1965 – 1985+)

- Interaction style
 - User issues a command, waits for response
 - Feedback can be given during execution
 - Commands need to be learned
 - Commands are hard to discover
- Users
 - trained experts

```
[mkyong@localhost _node]$ du -lsh pattern_final
2.4G    pattern_final
[mkyong@localhost _node]$ du -lsh pattern3
726M    pattern3
[mkyong@localhost _node]$
```



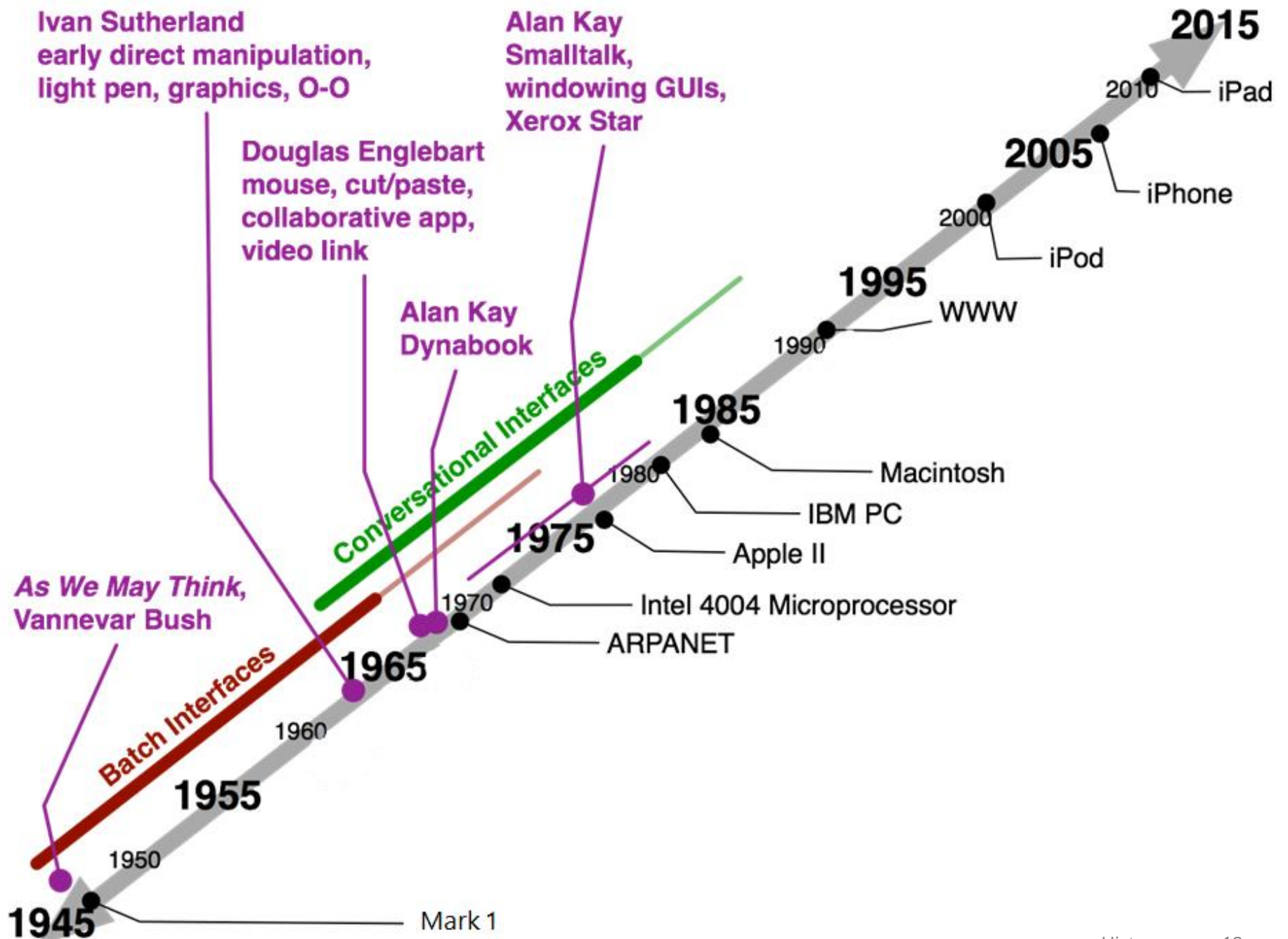
Command-Line Interface

- Advantages
 - Fast and highly flexible: can combine commands to create sophisticated sets of operations
- Disadvantages
 - Users need to understand the computer
 - I/O is in system language, not task language
 - Requires “Recall” rather than “Recognition”

Recognizing User Needs

- Batch and command line interfaces require interaction language closer to the system than task
- These interfaces were common at a time when the computer's time was more expensive than a person's time
- Some visionaries imagined a different form of interaction.





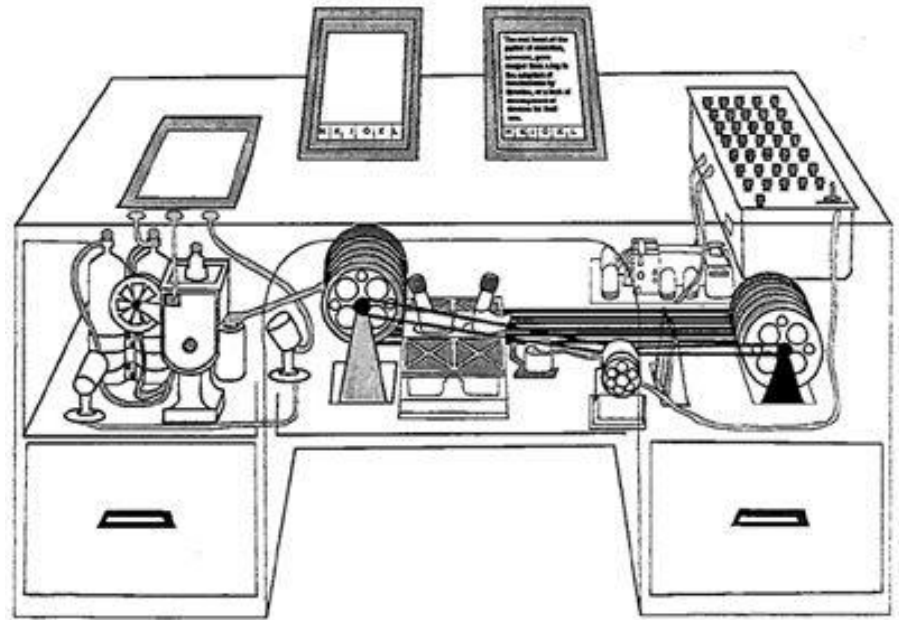
Vannevar Bush

- Headed Office of Scientific Research and Development
 - Manhattan project, other WWII science efforts
- 1945 article, “As We May Think” in The Atlantic
 - <http://www.theatlantic.com/doc/194507/bush>
 - inspires computer scientists to present day
- Goal was to augment human intellect



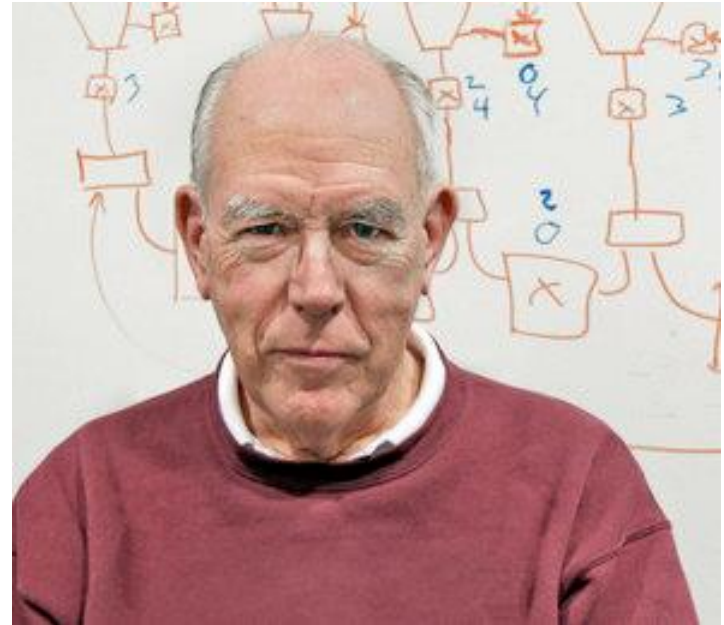
Bush's "Memex"

- "A memex is a device in which an individual stores all his books, records, and communications. It is an enlarged intimate supplement to his memory."
- Proposes associative links between content (hyperlinks)
- Dual display setup
- Direct annotation of stored content
- Indexing for easy access
- But hardware a long way off



Ivan Sutherland

- Sketchpad (~1963)
 - Light pen
 - Direct manipulation
 - Early graphical interface
 - <https://youtu.be/hB3jQKGrJo0?t=1m30s>
- Expands computer domain to include artists, draftsmen, ...
- Language of interface moves substantially closer to task domains



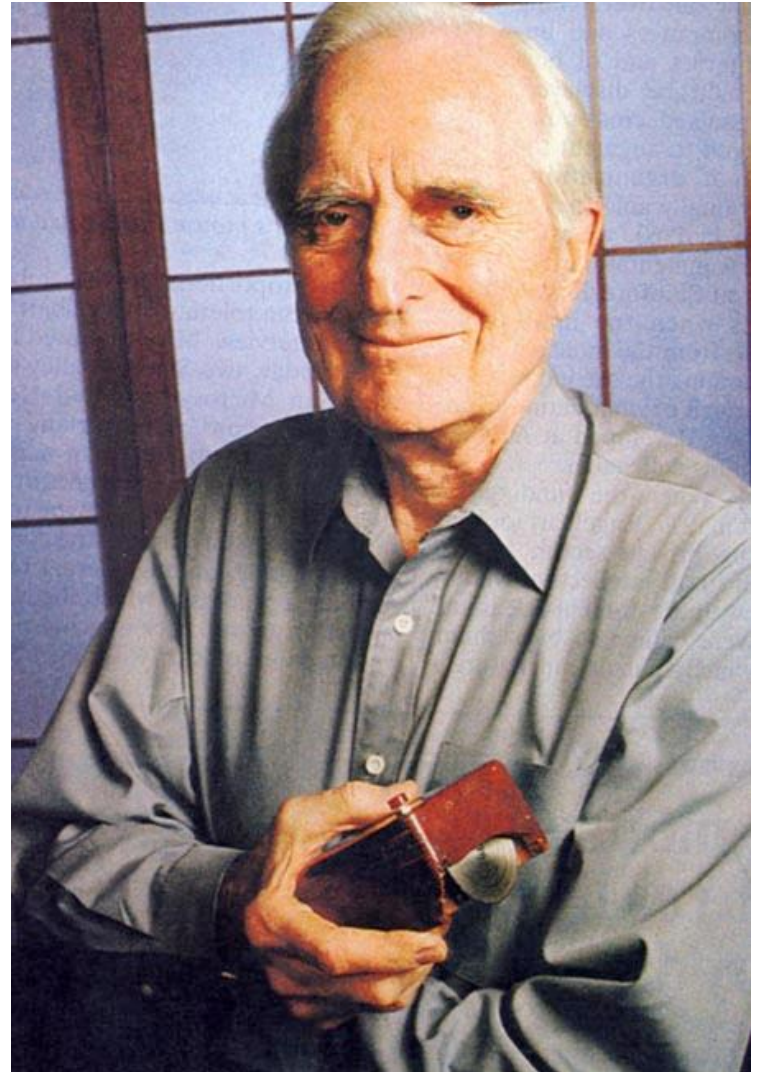
INK

+



Douglas Engelbart

- Led team at Stanford Research Institute (SRI) created On-Line System (NLS) (~1968)
 - invented the mouse
 - implemented hypertext
 - introduced copy/paste
 - vision of computer-supported collaborative work



The NLS “Mother of all Demos” (1968)



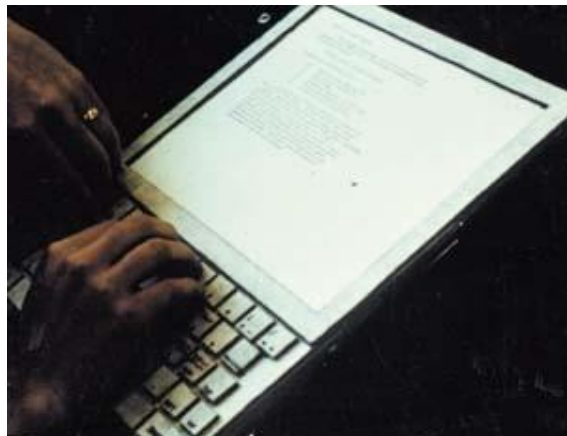
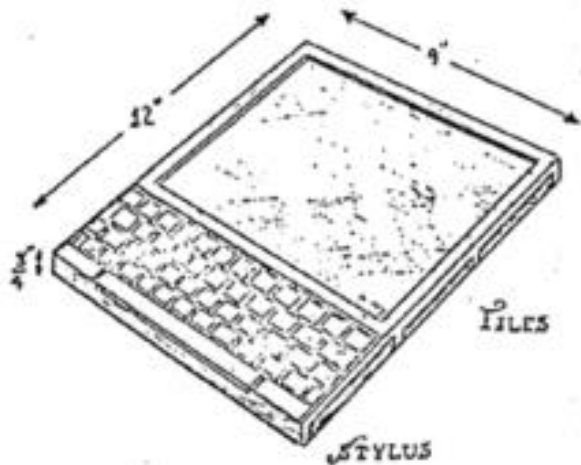
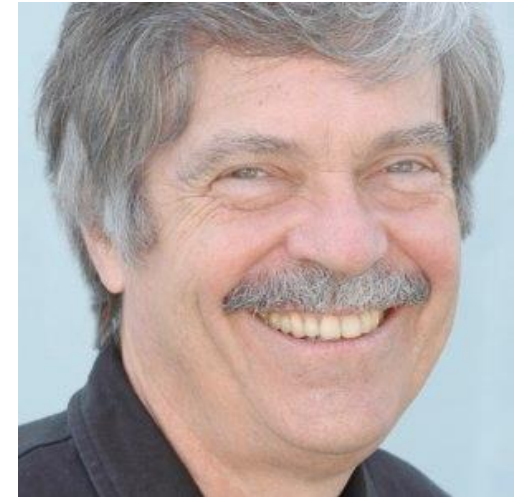
<https://youtu.be/yJDv-zdhzMY?t=90> (intro, text editing, copy & paste)

<https://youtu.be/yJDv-zdhzMY?t=1857> (mouse, chorded keyboard)

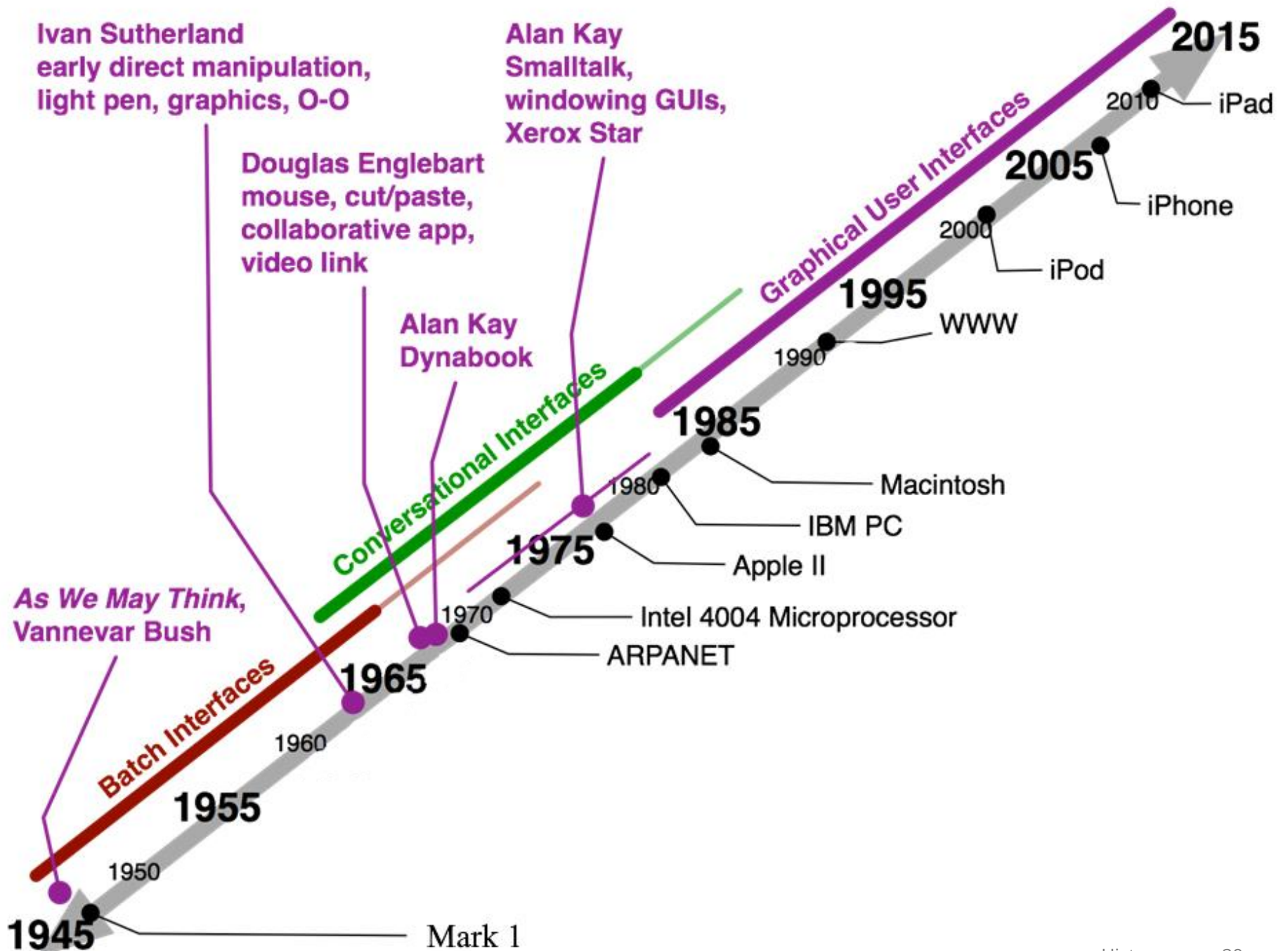
Talked about hardware, hypertext documentation, collaboration, mentions that Arpanet is coming)

Alan Kay

- Pioneering work on
 - object-oriented programming (Smalltalk)
 - Xerox Star: graphical user interface
 - Dynabook: conceptual basis for laptops and tablet computers
- Quote: "The best way to predict the future is to invent it."

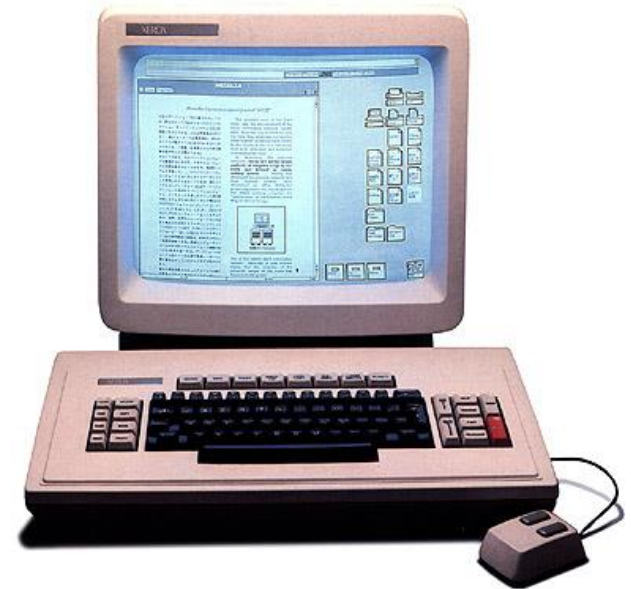
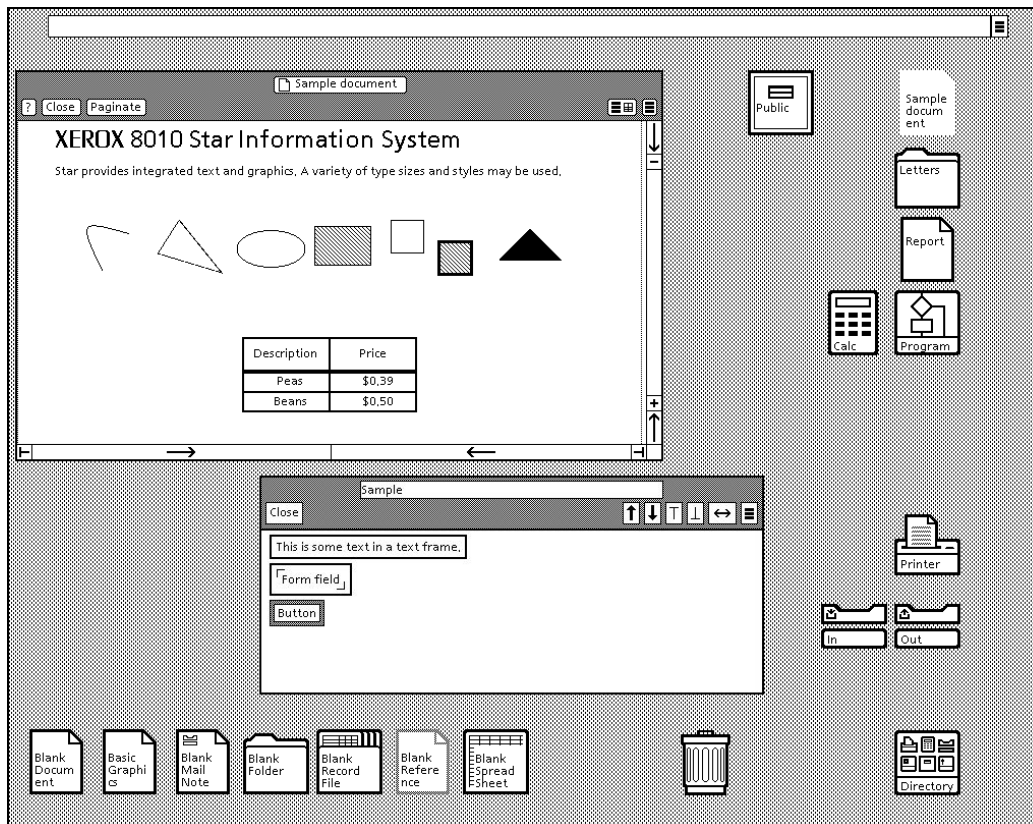


Dynabook (~1972)



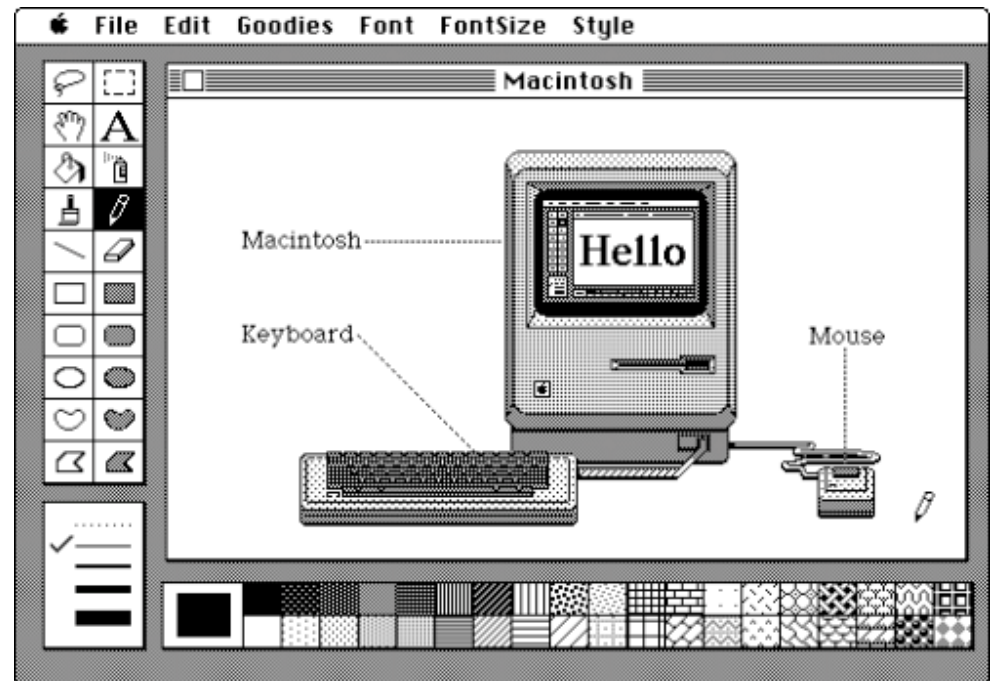
Xerox Star Information System (~1981)

- First commercial computer with GUI
 - windows, icons, folders, mouse, (and Ethernet, file/print servers, email)
 - based on Xerox Alto research ~1974

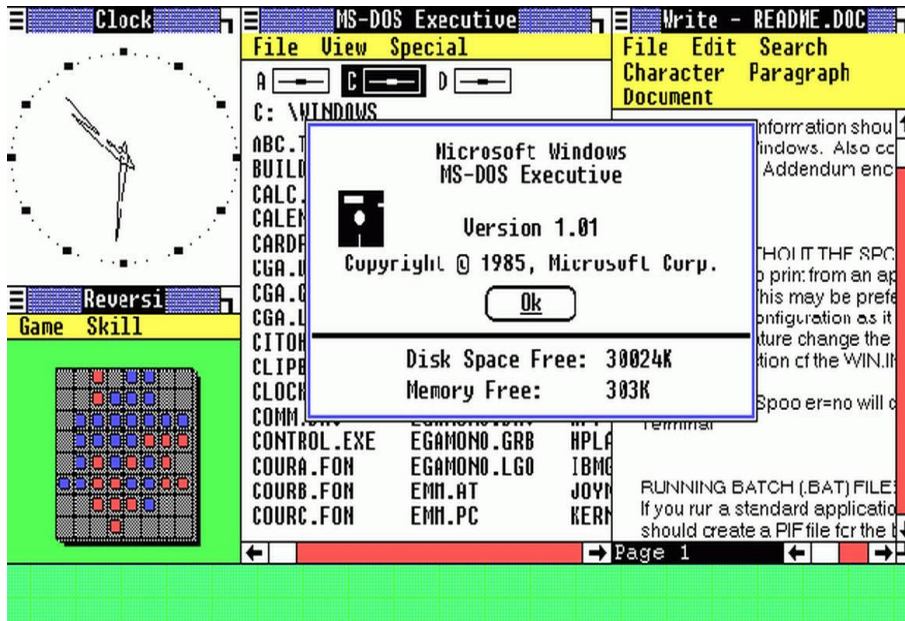


Graphical User Interfaces (1984 – present)

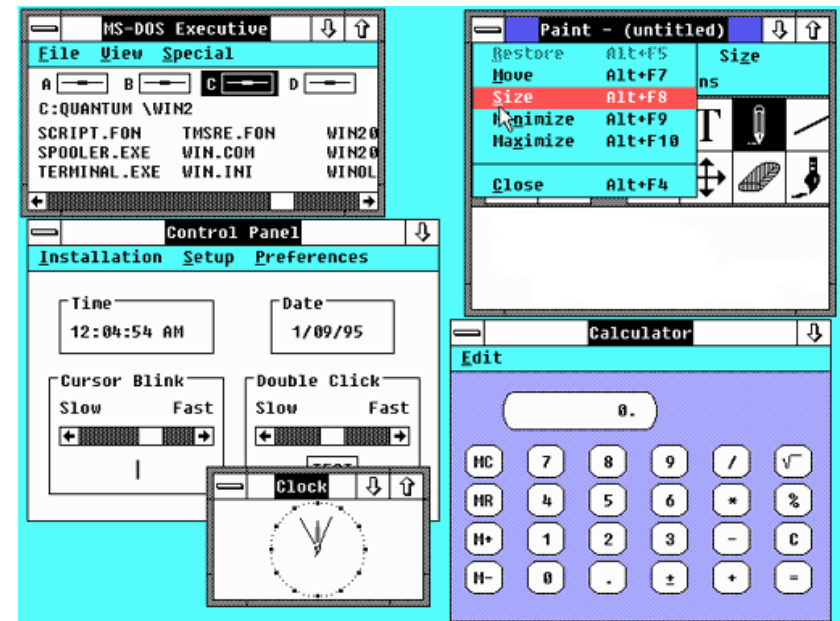
- The early 1980s, Apple became the world's most popular computer
- IBM enters the personal computer market in 1981 with the IBM PC
- Apple's Macintosh (Jan 1984), brings the GUI to the masses



Graphical User Interfaces (1984+)



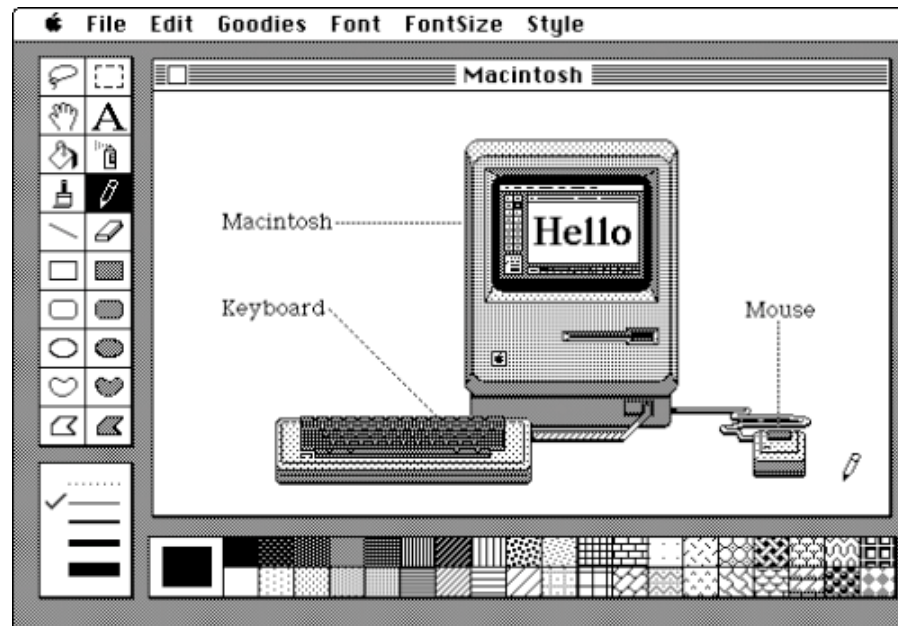
Microsoft Windows 1.0 (1985)



Microsoft Windows 2.0 (1987)

Graphical User Interface (GUI)

- WIMP interface
 - Windows, Icons, Menus, and Pointer
- Hardware interface
 - Keyboard
 - Pointing device (e.g., mouse)
 - High resolution, high refresh graphics display



GUI Interaction style

- User in control
 - system waits for input, then responds
- **Recognition** over **Recall**
 - enables discovery of options and experimentation
- Metaphors
 - make Interaction language closer to users' own language, closer to task domain
 - e.g. "desktop", "folder", "drag-and-drop",...
- GUI interaction opens interface up to broader audience

Modern and Future Interaction

- Where can we go from here?
- What other paradigms are possible?



Voice interface

- Voice also started to become a form of interaction with the 1997 introduction of Dragon NaturallySpeaking
 - sold several million copies of its voice interaction software
- Google Home
 - <https://www.youtube.com/watch?v=r0iLfAV0plg>
- Amazon Echo
 - <https://www.youtube.com/watch?v=zmhcPKKt7gw>



Touch interfaces

- The latter part of the 2000s witnessed a big leap forward in UI design
- FingerWorks, a gesture recognition company, produced multi-touch products in 1998
 - The company was acquired by Apple in 2005.
- Touch interfaces took off dramatically with the introduction of the iPhone in 2007 and the iPad in 2010
 - Multipoint capacity touch enabled users to interact with digital content in new ways



Gesture interface



“Kinect” (2010) <https://www.youtube.com/watch?v=pzfpXAbQ61U>



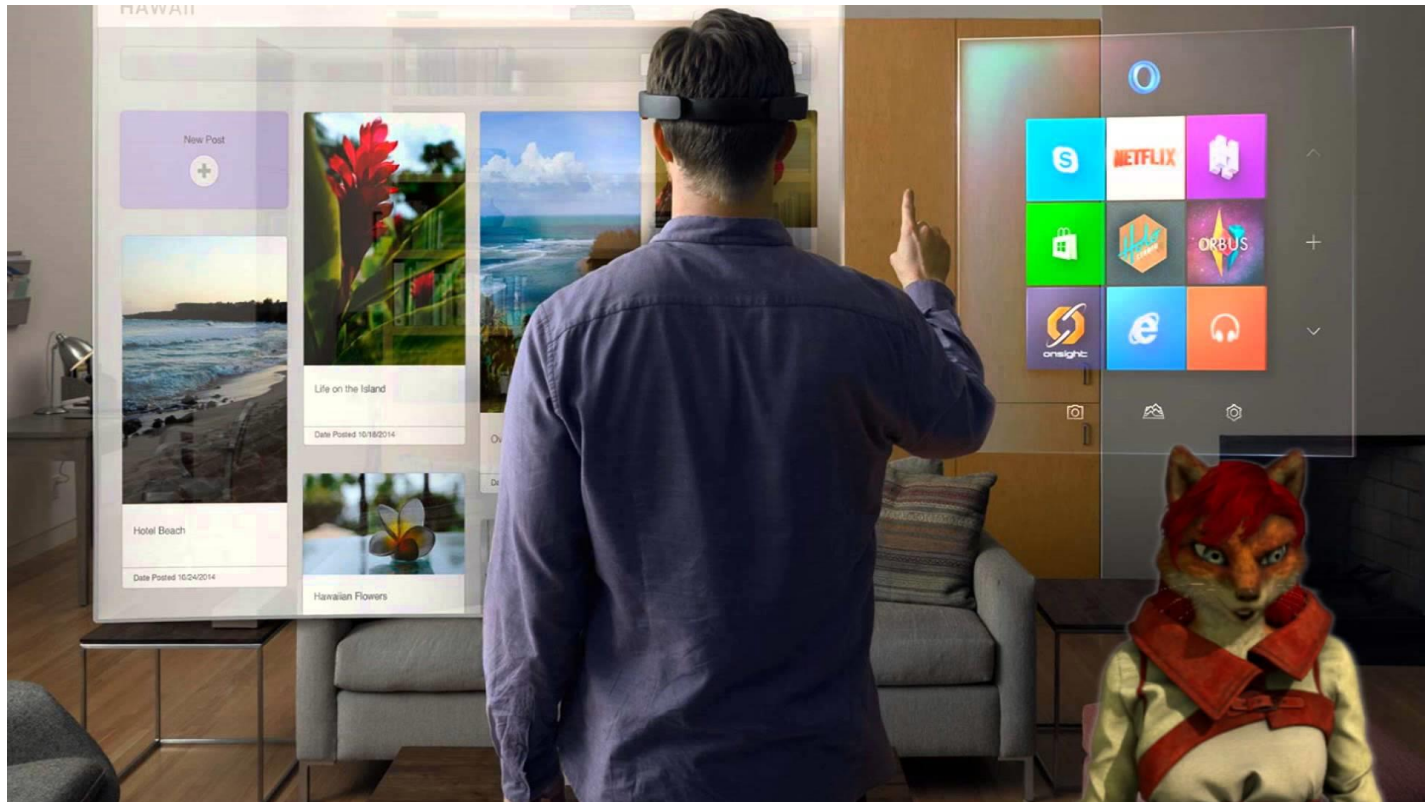
“MYO armband” (2013)
<https://youtu.be/jOEcsNmTk7g?t=157>

“Leap Motion” (2012) https://www.youtube.com/watch?v=_d6KuiutelA

Augmented Reality

- HoloLens 2 AR Headset (2016)

<https://www.youtube.com/watch?v=uIHPPtPBgHk>



Virtual Reality

- Oculus Quest 2 (Meta Quest Pro – latest release)
- <https://www.youtube.com/watch?v=60yP8f5E-B4>
- Apple Pro Vision:
https://www.youtube.com/watch?v=TX9qSaGXFyg&ab_channel=Apple



Summary

- History of HCI

	Input	Output
Early Days	Connecting wires Paper tape & punch Card Keyboard	Lights on display Paper
Today	Keyboard Mouse Microphone Cameras	High-res screen Audio Head mounted displays
Soon?	Data gloves + suites Computer jewelry	Ubiquitous computing Multimedia

Mark Wessier

- **“The most profound technologies are those that disappear.**
They weave themselves into the fabric of everyday life until they are indistinguishable from it.” (*Scientific American*, 1991)
- Computing should fade into the environment
 - examples?



Ubiquitous Computing

- A Day Made of Glass... Made possible by Corning. (2011)
- https://www.youtube.com/watch?v=6Cf7IL_eZ38



Microsoft Vision (2015)

- <https://www.youtube.com/watch?v=cEeht3nyhec>



History

A (brief) history of interaction